INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RANCHI

Minor Project Mid-Term Presentation

on

EXPLAINABLE ACCOUNTING-FRAUD DETECTION USING MACHINE LEARNING

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PROBLEM STATEMENT

- Rank publicly traded U.S. firms by their risk of accounting fraud, one year ahead, so that a limited investigation budget reaches the riskiest firms first.
- Fraud label: SEC Accounting and Auditing Enforcement Releases (AAERs).
- The model must stay explainable and use only information available at scoring time.

Firm-year i in fiscal year t: features $x_{i,t} \in \mathbb{R}^d$, label $y_{i,t} \in \{0, 1\}$

Learn a score $s : \mathbb{R}^d \to [0, 1]$ that ranks the firms of test year T

Training data: $\{ (x_{i,t}, y_{i,t}) : t \leq T - 2 \}$ (labels arrive late)

Goal: maximise $AUC = P(s(x^+) > s(x^-))$ and recall in the top k %

Base rate: $\pi = 964 / 146{,}045 \approx 0.66\ \%$

146,045
firm-years, 1990–2014

964
fraud firm-years

412
distinct fraud cases

0.66 %
base rate

INTRODUCTION

- **Accounting fraud:** deliberate misstatement of financial reports; costly and rarely visible from outside.
- **Labels:** SEC AAERs, issued years after the fraud, so recent years look cleaner than they are.
- **Data:** Compustat fundamentals, 1990–2014 in the public file of the base paper; the model uses 1991–2014 (1990 only supplies prior-year values).
- **Base paper:** Bao, Ke, Li, Yu & Zhang (2020), Journal of Accounting Research [1].
  - 28 raw financial items as inputs instead of expert ratios
  - RUSBoost: boosting with random under-sampling for rare events
  - Beats the F-score [2] and an SVM [3]; AUC ≈ 0.72 on 2003–2008

RUSBoost, round $m = 1 \ldots M$:

sample non-frauds so that $|\text{non-fraud}| = |\text{fraud}|$

fit a shallow tree h_m, update weights (AdaBoost)

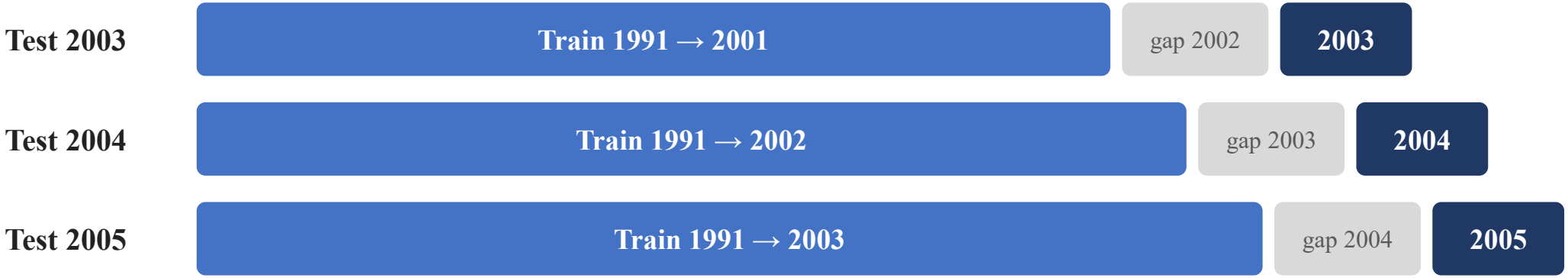
$F(x) = \Sigma_m \alpha_m h_m(x)$

28 raw items	**≈0.72** AUC (reported)	**2 yrs** train-test gap

Reported AUC is approximate; check against the paper before citing.

INTRODUCTION (CONTD.) | How models are tested

Expanding window, one row per test year



Expanding window

Retrain for every test year on all earlier data.

2-year gap

Frauds of year T − 1 are not public yet, so T − 1 is left out.

Serial-fraud masking

A fraud case that continues into the test year is removed from training.

$\text{Train} = \{ t \leq T - 2 \} \setminus \{ \text{cases active in } T \}, \quad \text{Test} = \{ t = T \}, \quad T = 2003, \ldots, 2012$

LITERATURE SURVEY

Study	Key Focus	Tools / Techniques	Findings
Altman [5], 1968	Bankruptcy prediction	Discriminant analysis on 5 ratios (Z-score)	Ratios separate distressed firms; Z-score is one of our features
Dechow et al. [6], 1995	Earnings management	Accrual models (Jones, modified Jones)	Modified Jones model had the most power of those tested
Beneish [4], 1999	Earnings manipulation	Probit on 8 ratio indices (M-score)	Indices flag manipulators; M-score parts are our features
Cecchini et al. [3], 2010	Management fraud	SVM with a financial kernel	Benchmark; about 0.63 AUC in the base paper's test
Seiffert et al. [10], 2010	Class imbalance	RUSBoost: under-sampling + boosting	Simple and fast; competitive with SMOTE-based boosting
Dechow et al. [2], 2011	Misstatements (AAERs)	Logistic F-score on financial ratios	Benchmark; about 0.69 AUC in the base paper's test
Perols [7], 2011	Comparing classifiers	LR, SVM, ANN, bagging, C4.5, stacking	Logistic regression and SVM performed best
Bao et al. [1], 2020 (base)	**AAER fraud prediction**	**RUSBoost on 28 raw financial items**	**Raw items beat expert ratios; about 0.72 AUC**

24 studies across three slides (the third covers 2024–2026); full references at the end.

LITERATURE SURVEY (CONTD.)

Study	Key Focus	Tools / Techniques	Findings
Brown et al. [8], 2020	Misreporting from text	Topic modelling (LDA) of annual reports	Text topics add signal to financial variables
Bertomeu et al. [9], 2021	Misstatement detection	Boosted trees; accounting, audit, market, governance data	Accounting data helps most combined with audit and market data
Chen & Guestrin [11], 2016	Scalable boosting	XGBoost: regularised, sparsity-aware trees	Strong tabular learner; a member of our ensemble
Ke et al. [12], 2017	Efficient boosting	LightGBM: histograms, GOSS, EFB	Much faster training at similar accuracy; a member here
Lundberg & Lee [13], 2017	Explaining predictions	SHAP (Shapley values)	One consistent framework for feature attribution
Lundberg et al. [14], 2020	Explaining tree models	TreeSHAP	Exact SHAP for tree ensembles; used for our explanations
Grinsztajn et al. [15], 2022	Tabular benchmark	Tree ensembles vs deep networks	Trees still beat deep learning on typical tabular data
Zhang et al. [16], 2023	Automated features	OpenFE: expand-and-reduce	Generated features rival expert ones; tested here, no gain

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LITERATURE SURVEY (CONTD.) | 2024–2026

Study	Key Focus	Tools / Techniques	Findings
Bhattacharya & Mićković [25], 2024	Fraud from MD&A text	Fine-tuned BERT on 10-K MD&A sections	Beats text and financial benchmarks by 15 % and 12 %
Hernandez Aros et al. [26], 2024	Survey of ML fraud detection	Literature review	Maps the ML methods and data used for financial fraud
Chen et al. [27], 2025	Deep learning for fraud	Systematic review of 57 studies, 2019–2024	Imbalanced data and interpretability remain open
Ketelaar & Mićković [28], 2025	Language of fraudulent filings	BERT on linguistic indicators of 10-Ks	Fraud filings: more positive words, inconsistent language, higher readability
Amin & Aßenmacher [29], 2025	Open benchmark data	Labelled 10-K MD&A data set; ML and DL baselines	First public data set of 10-K sections with fraud labels
Ergun et al. [30], 2025	LLMs for fraud	FinBERT, FinLlama, FinGPT embeddings of MD&A over time + LSTM	Beats word-frequency methods on 1995–2024 reports
Nie et al. [31], 2025	Multimodal, interpretable	GBDT on financial, governance and text features + SHAP	Governance and text added to financials; SHAP names key risk indicators
Sodnomdavaa & Lkhagvadorj [32], 2026	Explainable ML, emerging market	ML + SHAP/LIME + decision-curve analysis; 969 firm-years	Leverage, profitability and liquidity drive misstatement risk

Trend: text and LLMs, richer firm data, and explainability. Imbalance and interpretability, which this project addresses, remain open.

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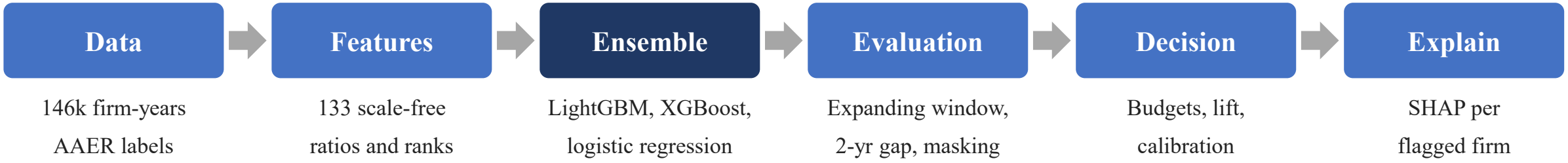
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CHALLENGES & LIMITATIONS

Gap in the base paper	Why it matters	How this project addresses it
1. Scale bias	Raw dollar items let firm size drive the score	Scale-free ratios, each ranked within its fiscal year
2. Black box	300 boosted trees give no reason for a flag	SHAP explanation for every flagged firm
3. No decision step	Only a ranking: no cutoff, no usable probability	Precision, recall and lift at review budgets; calibrated probability
4. Thin statistics	No confidence intervals, though cases span years	Paired bootstrap clustered by fraud case and firm
5. Not deployable	No way to score new firms	Deployable model and a scoring script

Extreme imbalance	Late labels	Dependent rows
Only 0.66 % of firm-years are fraud	Test-year frauds fall from 69 in 2003 to 4 in 2014	964 fraud firm-years come from only 412 cases

PROPOSED METHODOLOGY



Implementation modules (Python: pandas, LightGBM, XGBoost, scikit-learn, SHAP)

Module	What it does	Code
Feature engineering	Ratios, Beneish and Altman parts, growth gaps, within-year ranks; winsorising refit per window	src/features.py
Models	Balanced-bagged LightGBM and XGBoost, logistic regression, RUSBoost baseline	src/models.py
Evaluation	Yearly AUC, NDCG, recall and precision at k; clustered bootstrap	src/evaluate.py
Explanation	TreeExplainer on every bag, permutation SHAP for logistic regression	src/explain.py
Pipeline / deploy	End-to-end run (about 2 h on a laptop CPU); scorer for new firms	run_pipeline.py, score_new_data.py

PROPOSED METHODOLOGY | Feature engineering

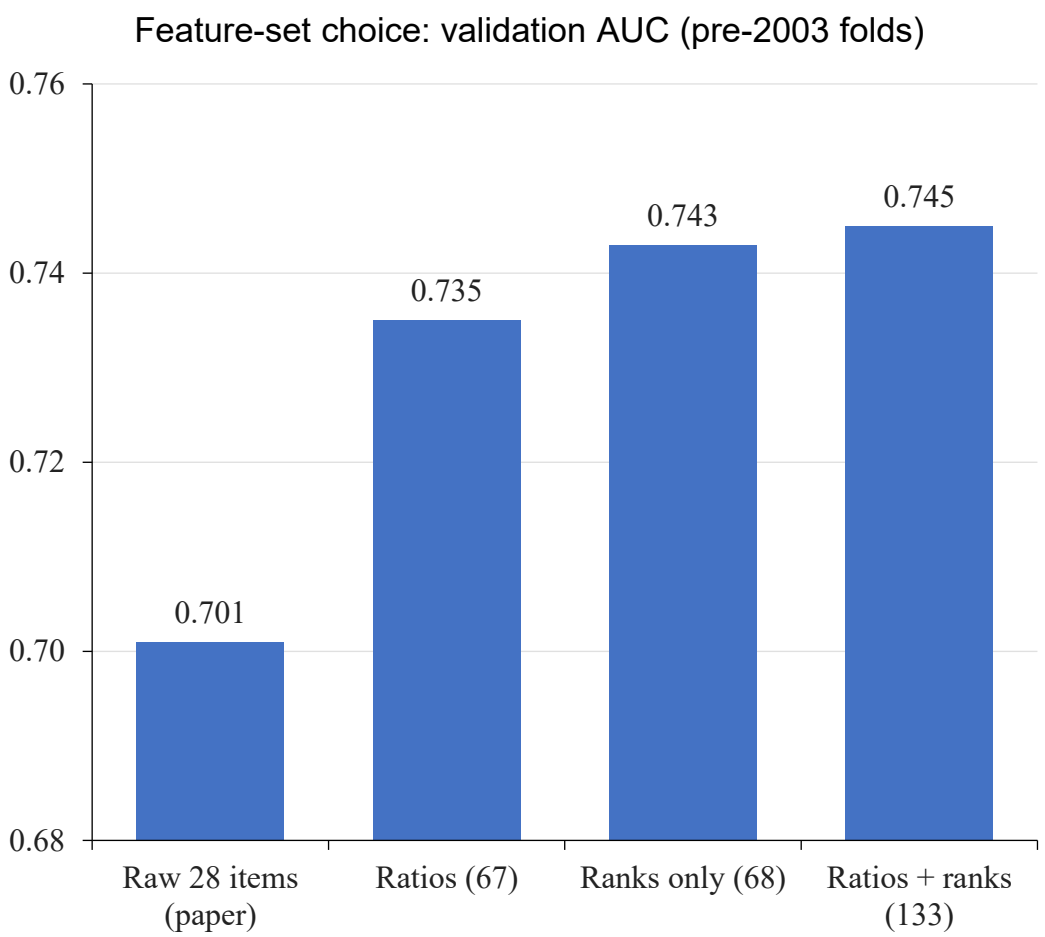
- **Removed:** the 28 raw dollar items (kept only for the baseline).
- **Added:** 14 paper ratios, Beneish M-score parts, Altman Z, common-size ratios, growth gaps, financing need (67).
- **Ranked:** each ratio as a percentile within its fiscal year (+66).
- **Cleaned:** clipped at the 1st/99th percentile of each training window only.

$$DSRI = (REC_t / SALES_t) / (REC_{t-1} / SALES_{t-1})$$

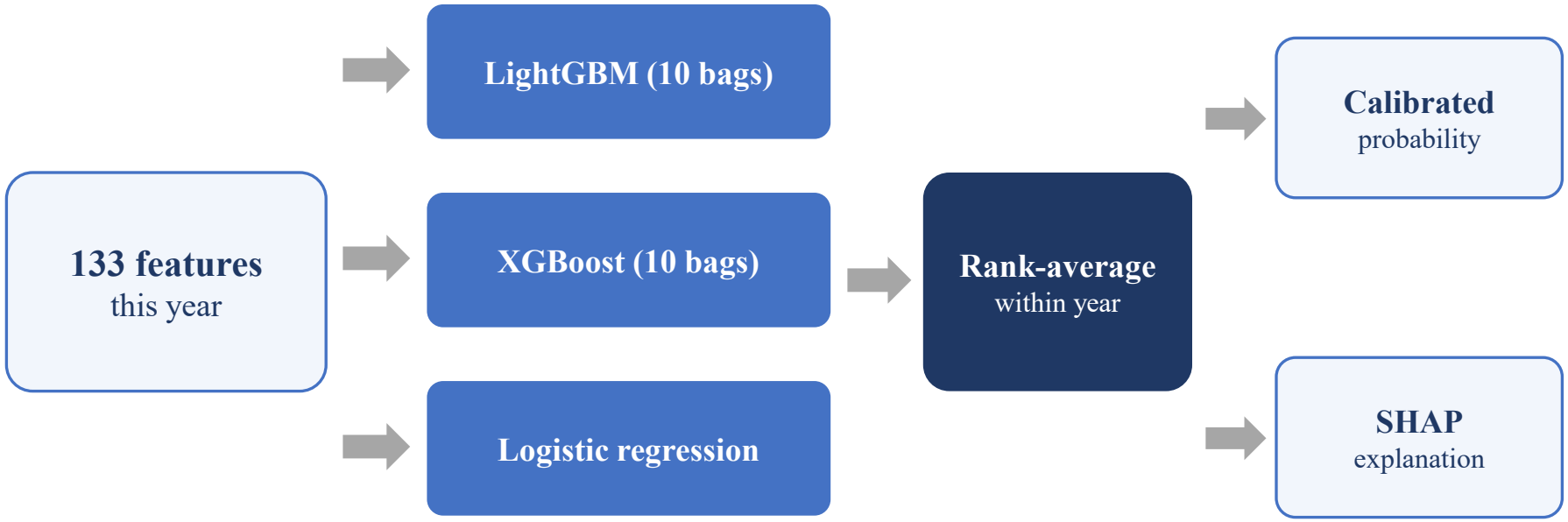
$$Z = 1.2 \cdot X_1 + 1.4 \cdot X_2 + 3.3 \cdot X_3 + 0.6 \cdot X_4 + 1.0 \cdot X_5$$

$$rank_t(f)_i = (\text{position of } f_i \text{ among firms of year t}) / n_t$$

Total: 67 ratios + 66 within-year ranks = 133 features



PROPOSED METHODOLOGY | Ensemble and explanation



$p_m(x) = (1/B)\ \Sigma\ p_m^{(b)}(x),\ B = 10$ bags

bag b: all frauds + $\rho \cdot |F|$ non-frauds

$s(x) = \frac{1}{3}\ \Sigma_m\ \mathrm{rank}_t(p_m(x))\ /\ n_t$

$\hat{P}(\mathrm{fraud} \mid s) = g(s),\ g$ isotonic $\uparrow$

$f(x) = \varphi_0 + \Sigma_j\ \varphi_j(x)$ (SHAP)

- **Imbalance:** balanced bags ($\rho = 5$ non-frauds per fraud for LightGBM, 20 for XGBoost) instead of discarding data.
- **Tuning:** random search, 20 trials per tree model, on folds that validate on 1999, 2000 and 2001 only.
- **Decision:** flag the top 1 %, 2 % or 5 % of each year; isotonic calibration fit on out-of-sample scores.

EXPERIMENTS | Setup and metrics

Setup

- Validation: rolling folds scoring 1999, 2000, 2001 (data before 2003 only)
- Test: 2003–2012, each year scored by a model trained on $t \leq T - 2$
- 2013–2014 excluded from averages (11 and 4 frauds)
- Candidate experiments repeated on 3 seeds (42, 43, 44)
- Deployable model trained on 1991–2010 (later labels incomplete)

Significance

- Paired bootstrap, 1,000 resamples [20]: frauds resampled by AAER case, non-frauds by firm
- Holm correction across candidates [21]

$$AUC = (1 / n^{+}n^{-}) \Sigma_i \Sigma_j 1[s(x_i^{+}) > s(x_j^{-})] \quad [24]$$

Top k % of a year with n firms: $K = [k \cdot n]$

$Sens@k = TP_k / n^{+}$ (recall in the top k)

$Prec@k = TP_k / K$

$Lift@k = Prec@k / \pi$

$$DCG@k = \Sigma_{j=1}^{K} y_{(j)} / \log_2(j + 1)$$

$NDCG@k = DCG@k / IDCG@k$ [22]

Gain Δ = metric(model) − metric(baseline),

95 % CI from the bootstrap percentiles

EXPERIMENTS | Rules and runs

Rules written first → **Validation gate** → **Adoption rule**

Candidates, settings and thresholds fixed before any run

≥ +0.002 validation AUC on fresh seeds

Significant ($p \leq 0.05$), positive on all seeds and in 2003–08 and 2009–12, no metric harmed

Experiment	What was tested	Outcome
E1 Rebuild of the base paper	RUSBoost from the authors' settings, same data and protocol	Reproduced
E2 Feature-set selection	Raw 28 vs ratios vs ranks vs ratios + ranks	133 features chosen
E3 Ensemble selection	LightGBM, XGBoost, LR, RUSBoost combinations	LGBM + XGB + LR
E4 Dynamic feature selection	10 families inside every training window	Rejected
E5 Automatic feature generation	OpenFE-style expand-and-reduce, 1 and 2 rounds	Rejected
E6 Other learners and losses	Dynamic RFE, focal PU loss, TabM, Mambular	Rejected
E7 Filing-text data	Text tone, industry, timing, amendment features	Rejected
E8 Evaluation-rule diagnostic	Same model under weaker test rules	Explains higher AUCs

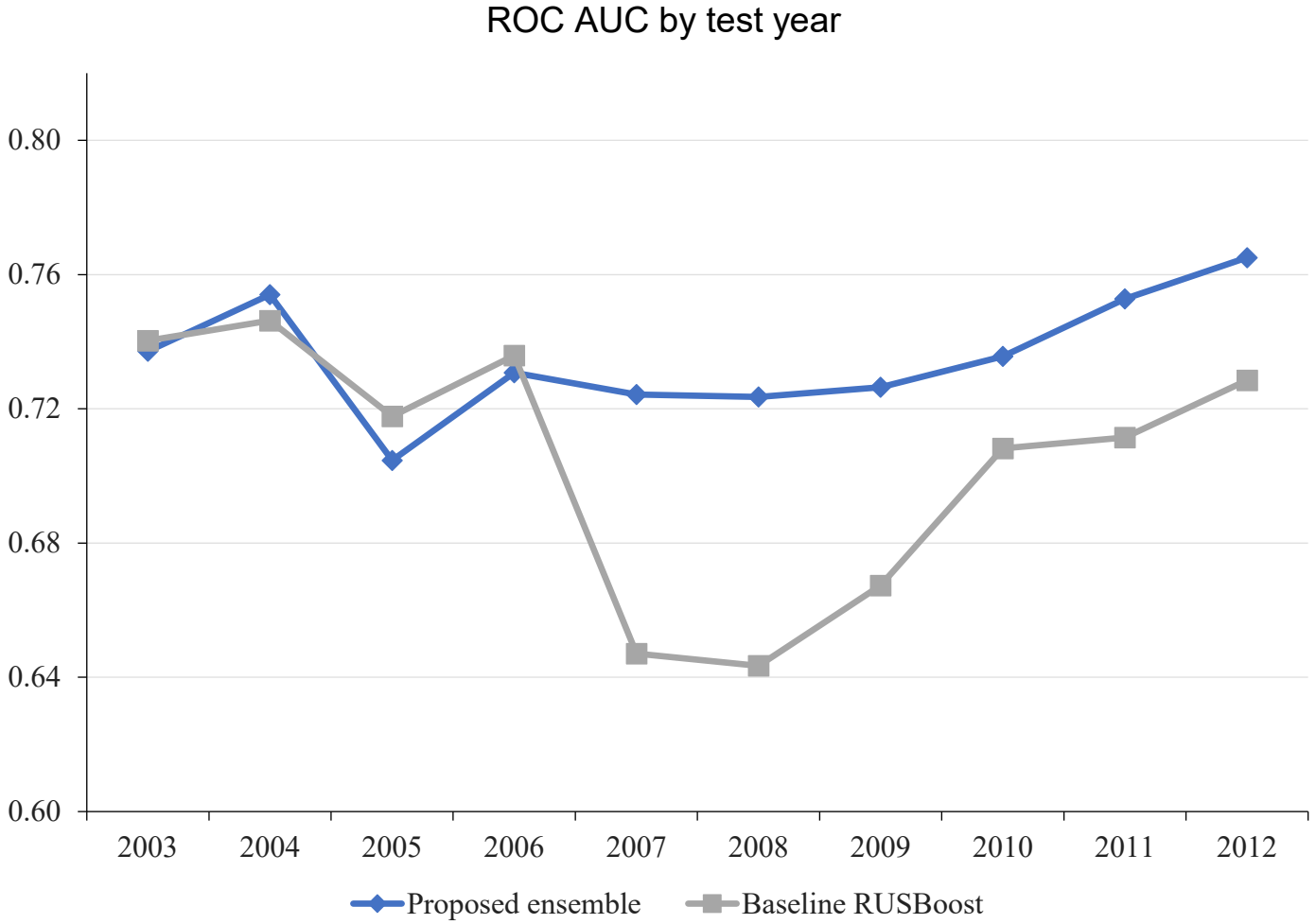
RESULTS | Overall performance

Model (test years 2003–2012)	AUC	NDCG@5%	Sens@5%	Prec@1%	Lift@1%
Baseline: RUSBoost, raw 28 items [1]	0.705	0.066	9.8%	1.8%	3.1
RUSBoost, our 133 features	0.732	0.071	12.7%	1.6%	2.7
LightGBM, balanced bags	0.730	0.074	14.2%	1.9%	3.5
XGBoost, balanced bags	0.726	0.072	14.2%	1.6%	2.7
Logistic regression	0.725	0.079	15.6%	1.4%	2.1
Proposed: LightGBM + XGBoost + LR	**0.735**	**0.082**	**15.6%**	**1.7%**	**3.3**

Gain of the proposed model over the baseline	Gain	95 % CI	P(gain ≤ 0)
AUC	+0.031	[+0.011, +0.049]	0.001
NDCG@5%	+0.015	[−0.011, +0.040]	0.133
Sens@5%	+0.058	[+0.015, +0.096]	0.002

Paired bootstrap, 1,000 resamples, clustered by AAER case and firm. 2013–2014 excluded (fewer than 15 frauds).

RESULTS | By year and review budget



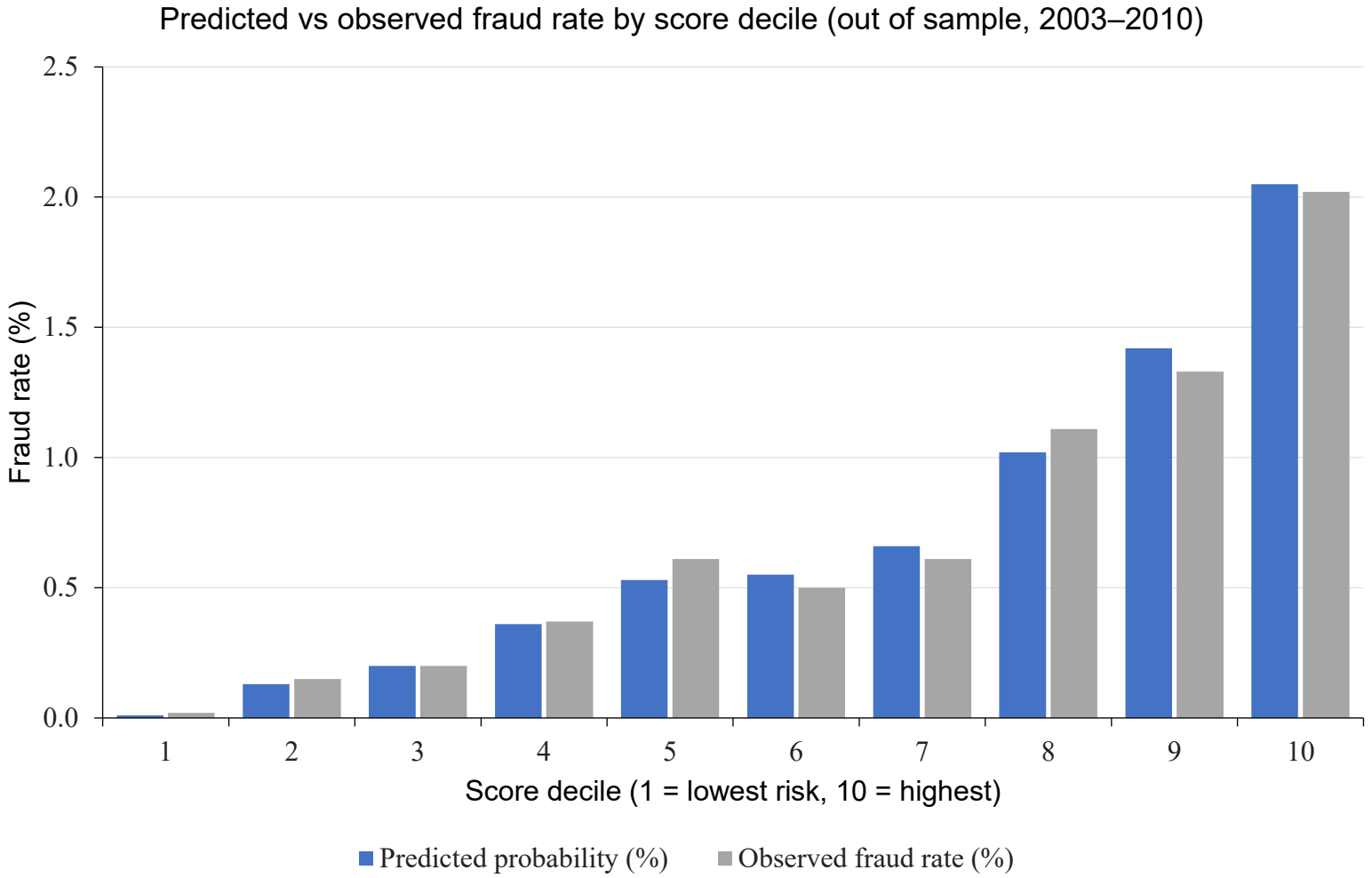
Budget	Precision ours	Precision base	Recall ours	Recall base
Top 1%	1.7%	1.8%	3.4%	3.2%
Top 2%	1.8%	1.6%	5.8%	4.4%
Top 5%	2.0%	1.3%	15.6%	9.8%

15.6%
of all frauds caught in the top 5 % (baseline 9.8%)

At a 1 % budget: 1.7 frauds per 100 reviews (baseline 1.8); random picking finds about 0.6.

Mean of yearly values, 2003–2012.

RESULTS | Calibrated fraud probability

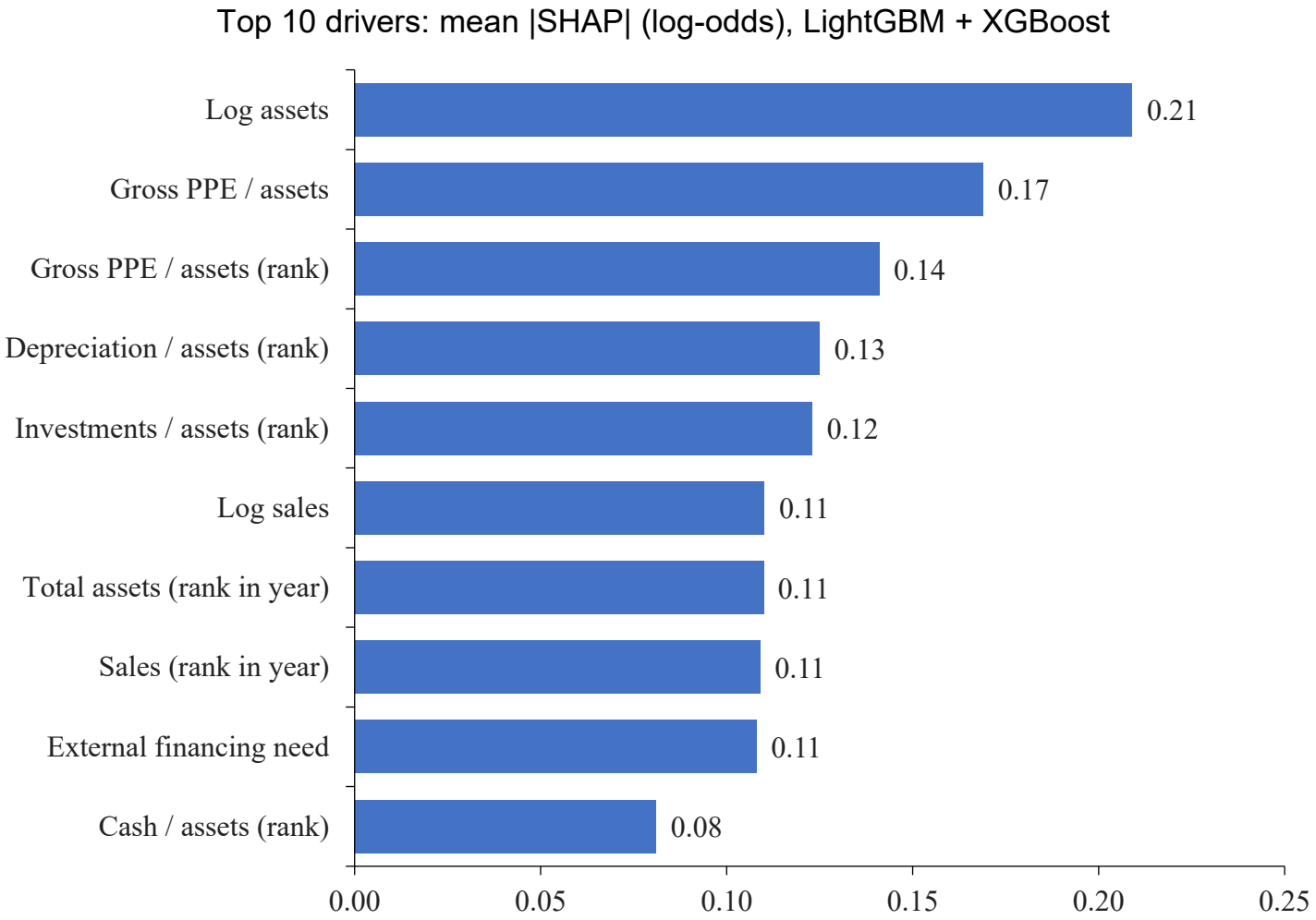


How to read it

- Top decile: predicted 2.05 %, observed 2.02 %.
- Bottom decile: predicted 0.01 %, observed 0.02 %.
- The top tenth of firms is about 90 times riskier than the bottom tenth.
- Fitted on the same scores, so agreement is partly by construction; it saturates near 2–3 %.

Isotonic regression [23]; pipeline output results/calibration.csv.

RESULTS | Explainability (SHAP)



How to read it

- Each bar is the average size of a feature's push on the fraud score.
- Drivers an auditor recognises: fixed assets, depreciation, investments, financing need, cash.
- Firm size still ranks first, even without raw dollar items.
- Per firm, a waterfall shows which features raised or lowered its score.

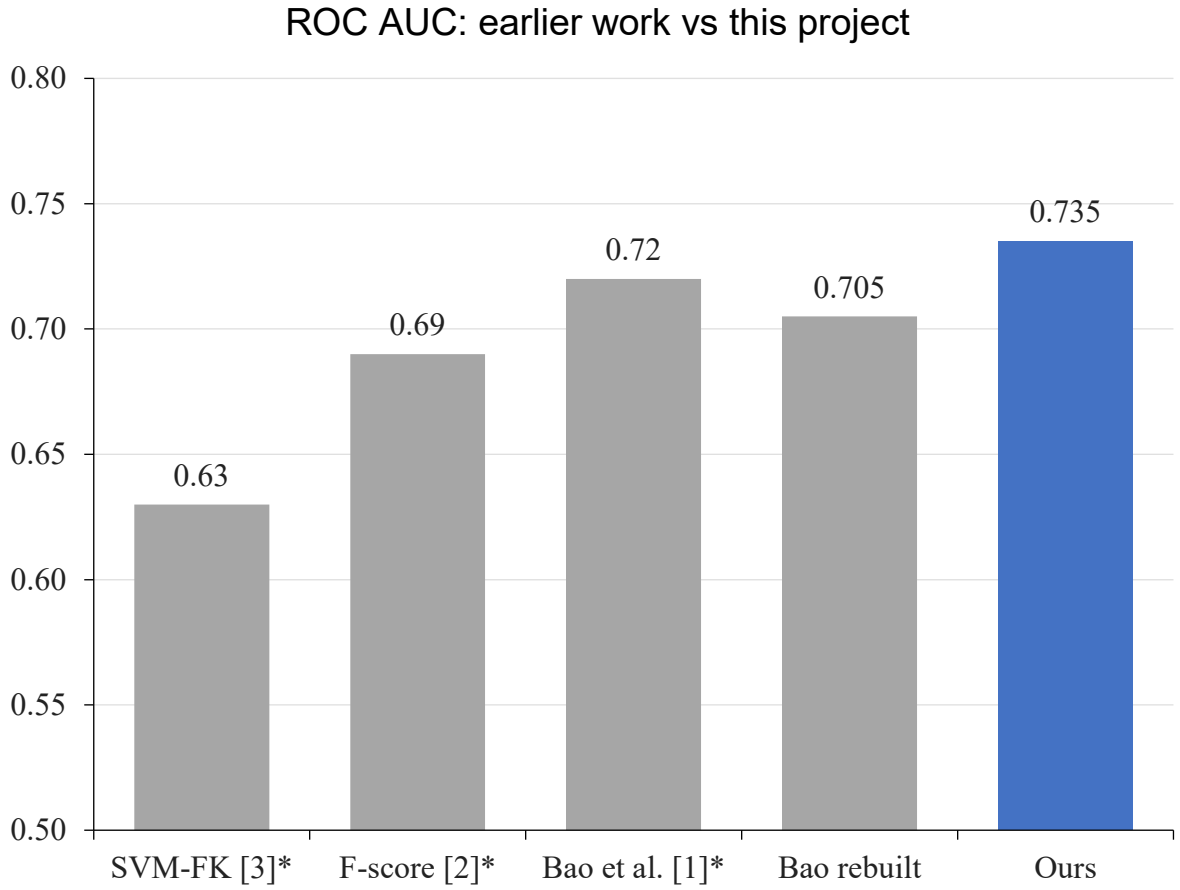
SHAP shows what the model uses, not why a firm committed fraud.

RESULTS | Fresh-period test, fiscal 2015–2018

Fresh test, SEC data	Firm-years / frauds	AUC	95% CI
Fiscal 2015–2017, mean of yearly AUCs (primary)	**12,474 / 55**	**0.776**	**0.721–0.828**
Fiscal 2015–2018 pooled	16,478 / 67	0.769	0.711–0.820
Only cases the SEC calls fraud	16,449 / 38	0.773	0.689–0.836
Only filer-years with a price	14,086 / 60	0.766	0.712–0.815
Only total assets of at least $10m	12,310 / 64	0.747	0.697–0.799
Fiscal 2019–2020 (labels incomplete)	7,974 / 8	0.779	0.637–0.939

The saved model (trained on fiscal 1991–2010) was scored once, unchanged, on SEC XBRL financials and new hand-built fraud labels, none of it in the paper's file. The same model scores 0.701 on Compustat and 0.700 on SEC inputs for fiscal 2012–2014, so the data source costs nothing. The pre-set bar was 0.725.

DISCUSSION | Comparative analysis



** As reported in the base paper (2003–2008), approximate. Ours and the rebuild: 2003–2012.*

Where the gain comes from

- **+0.0275** from our scale-free features alone (same RUSBoost)
- **+0.0033** from the three-model ensemble

Why the features are fixed, not dynamic

- Chosen once, as a set, on validation years; the test years never influenced it.
- 10 selection families and 2 generation methods ran inside every training window; none cleared the +0.002 bar.
- Directly on the test years, dynamic selection changed AUC by +0.0008 (95 % CI −0.0043 to +0.0062): no measurable difference.
- For the same year it kept 59 to 200 features depending only on the seed. Fixed features keep SHAP comparable across years.

DISCUSSION | Limits and why not 0.8

Same model, only the evaluation rule changes

Evaluation rule	Change in AUC
Our rule: window, 2-yr gap, masking	reference
Without serial-fraud masking	+0.047
Without masking and without the gap	+0.087
Random 5-fold split by firm	+0.036
Random 5-fold split of firm-years	+0.080

Weaker rules keep labels a real user cannot have yet (AAERs arrive years later), or put one fraud case on both sides of the split. We do not claim 0.8.

Limitations

- **A ranking tool, not a detector:** most flagged firms are not labelled fraud.
- **Firm size still matters:** sales and assets rank highest in SHAP.
- **Incomplete labels:** only SEC-caught fraud is labelled; recent years are thin.
- **Small validation set:** 3 validation years, about 300 frauds; differences near 0.005 are noise.
- **Reused test years:** later experiments are exploratory, not confirmatory.

CONCLUSION AND FUTURE WORK

Conclusion

- Reproduced the base paper's RUSBoost in Python, in its reported range.
- Scale-free features + ensemble: AUC 0.705 → 0.735; top-5 % recall 9.8% → 15.6% (both significant).
- Added review budgets, calibrated probabilities, clustered significance tests and SHAP explanations.
- Showed that feature selection, feature generation and deep tabular models do not help on this data.
- Delivered a deployable model and a scoring script for new firms.

Future work

Firm history

A Temporal Transformer that reads each firm's last 5 fiscal years, since fraud builds up over several years.

Market signals

Within-year ranks of market-value and share-count movement, with recent training years weighted more [9].

Both will be tested under the same rules: validation gate first, then the adoption rule on the test years.

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THANK YOU